

Venkata Saratchandra Patnaik Markonda Patnaikuni

+1 6023018122 | smarkond@asu.edu | <https://www.linkedin.com/in/saratchandra-92015>

EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

Bachelor of Technology in Computer Science (CGPA: 9.11/10)

May 2022

Christ University School of Engineering, Bengaluru, India

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Engineered real-time streaming workflows for WebRTC and WebRTS clients, converting and delivering live media streams with optimized codec compatibility and low latency to diverse playout platforms.
- Evaluated and selected media codecs (for example, H.264/H.265 variants) and AWS MediaLive configurations to maximize clarity and compatibility across devices, reducing client playback errors by 95%.
- Collaborated with broadcast and streaming teams to integrate WebRTC/WebRTS protocols, device input (HID/Xinput) support, and client-platform optimizations across environments.
- Managed the CI/CD lifecycle of over 15 containerized microservices on AWS EKS using Docker & Kubernetes; streamlined deployment and infrastructure as code (Terraform, Python) to cut manual setup by 90%.

PROJECTS

Parallel Implementation of Gaussian Filter on GPU (CUDA, PyCUDA, Image Processing)

- Implemented 2D Gaussian convolution on GPU using CUDA & PyCUDA, optimizing performance with thread grids, blocks, and shared memory.
- Applied techniques like loop unrolling, tiling, and separable filters, achieving up to 20× speedup over a CPU-based MATLAB implementation.
- Validated results using PSNR and SSIM metrics; analyzed computation time for various kernel/image sizes to quantify performance gain.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Built an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda. Automated deployments via shell scripts and passed over 100+ real-time autograder test cases.

TECHNICAL SKILLS

GPU & Parallel Computing: CUDA Programming, PyCUDA, Shared Memory Optimization, Loop Unrolling, Tiling, Grid-Block-Thread GPU Architecture

Programming & Scripting Languages: Python, C, C++, R, Java, Kotlin, Go, JavaScript, TypeScript

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Secure Networking & Linux Penetration – ASU Graduate Coursework

- Simulated TCP/IP client-server communications and packet interception using Python, netcat, and tcpdump.
- Emulated BGP/ARP attacks and crafted raw Ethernet/IP/TCP packets with Scapy to explore network security flaws.
- Performed deep inspection of Linux networking interfaces, routing tables, and firewall rules in a secure sandbox.

Relevant Coursework

Computer Organization and Architecture, System Software, Operating Systems, Computer Networks